

NISHANTH TALLURI

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SUMMARY

Data Scientist and Generative AI Engineer with several years of experience delivering scalable production ML and RAG solutions for enterprise data. Designed and deployed models that reduced manual document review times and improved compliance interventions. Expert in NLP, LLM integration, data pipelines, and production-grade model deployment, with strong Python and cloud development skills.

TECHNICAL SKILLS

- **Data Science & Analytics:** Statistical Modeling, Predictive Analytics, Natural Language Processing (NLP), EDA, Feature Engineering, Experimentation & A/B Testing
- **Databases:** Oracle, Google BigQuery
- **Machine Learning & AI:** Scikit-learn, TensorFlow, PyTorch, Hugging Face Transformers, Deep Learning, ML Pipelines, Model Deployment, Model Optimization, Vertex AI
- **Development Tools:** Jupyter Notebooks, Git, Streamlit, AWS SageMaker, FastAPI
- **Cloud & Big Data Tools:** Google Cloud Platform (BigQuery, Vertex AI, Cloud DataProc), PySpark, Cloud-based ML Deployment, Docker
- **Programming & Data Handling:** Python, SQL, Pandas, NumPy, REST API Integration
- **Visualization Tools:** Tableau, Power BI, Seaborn, Plotly
- **Data Pipelines & Workflow:** Apache Airflow, Data Warehousing, ETL Pipeline Development, Feature Engineering
- **Generative AI & LLM Engineering:** Retrieval-Augmented Generation (RAG), LLM Application & RAG Pipeline Development, Embeddings & Semantic Search, Prompt Engineering & Optimization, LangChain, Vector Databases (Pinecone / FAISS / ChromaDB), Context Window & Token Optimization, LLM Evaluation & Monitoring, Inference Optimization

PROFESSIONAL EXPERIENCE

Cardinal Health

Aug 2024 - Present

Data Scientist

USA

- Built scalable ETL and feature engineering pipelines using Apache Airflow and BigQuery to support downstream ML and LLM applications, reducing manual data preparation time from 6 hours/week to 2.5 hours.
- Developed and deployed machine learning models using Scikit-learn and optimized data workflows for production, enabling high-risk pharmaceutical product identification and improving early compliance intervention across 6+ business teams.
- Built NLP and LLM-assisted document processing pipelines incorporating semantic chunking, embeddings generation, and topic modeling to analyze 100K+ regulatory documents and accelerate compliance reporting cycles.
- Deployed Retrieval-Augmented Generation (RAG) and AI-powered document intelligence solutions in cloud production environments using OpenAI APIs, Vertex AI, and Docker; automated regulatory review with robust integration and monitoring.
- Integrated LLM workflows with Dockerized data pipelines and Google Cloud services (BigQuery, Vertex AI) to deploy AI-driven document intelligence solutions in production, resulting in reliable, automated processing of regulatory documents and continuous compliance monitoring
- Optimized LLM response latency through prompt compression, token limit tuning, and embedding reuse strategies, improving overall response latency by ~30%.
- Developed executive dashboards using Power BI and Tableau to monitor model outcomes, adoption metrics, and risk indicators, enabling data-driven compliance decisions.

Tata Consultancy Services

May 2021 - Dec 2022

Data Scientist

India

- Developed demand forecasting models using Scikit-learn to support inventory planning across 25+ product lines handling over 1.2M monthly transactions, improving allocation efficiency and reducing quarterly overstock by 15,000 units.
- Migrated 30+ legacy SQL data pipelines to Google BigQuery and optimized distributed processing using Cloud Dataproc, reducing processing latency by 35% and lowering cloud storage costs by \$4,000 per month.
- Designed and maintained automated ETL workflows using Apache Airflow and Python to ensure robust data ingestion, feature engineering, and reproducible analytics for near real-time business reporting.
- Developed NLP-based text classification models for automated routing of 100K+ monthly customer feedback records, reducing manual triage effort by 25 hours per week and improving resolution efficiency.
- Processed and transformed 500GB+ transaction datasets using PySpark to improve model training efficiency and enable scalable analytics workflows.
- Built Seaborn and Plotly dashboards to monitor business KPIs and model performance trends, enabling early identification of data anomalies and model drift.

Shopify

Apr 2020 - Apr 2021

Jr. Data Scientist

India

- Performed exploratory data analysis on behavioral datasets from 800K+ users and developed churn prediction models using Scikit-learn, improving early churn detection accuracy by 18% and enabling targeted customer retention strategies.
- Built K-Means clustering pipelines to segment over 600K customer profiles, enabling targeted marketing campaigns and improving campaign ROI by 9%.
- Created Tableau dashboards using SQL-derived metrics to visualize churn trends and customer lifetime value across cohorts, helping product and marketing teams make informed decisions
- Developed internal Streamlit applications enabling non-technical stakeholders to generate churn predictions, reducing analyst dependency by 40%.
- Integrated and cleaned structured and unstructured datasets from multiple sources using Python and SQL, improving feature quality and model reliability.
- Conducted A/B testing experiments across three product features to evaluate impact on conversion and retention metrics within a 95% confidence interval.

EDUCATION

University of North Texas

Jan 2023 - Dec 2024

Master of Science, Computer Science

KLEF Deemed to be University

May 2017 - Apr 2021

Bachelor of Technology, Computer Science

PROJECTS

Customer Churn Prediction with NLP-Enhanced Feedback

- Designed an end-to-end churn prediction pipeline integrating NPS feedback sentiment analysis using TextBlob and structured behavioral data.
- Achieved an F1-score of 0.82+ using Random Forest and Logistic Regression, with feedback polarity labeled as Positive, Negative, or Neutral.
- Deployed model with Streamlit UI and visualized key sentiment patterns and churn drivers in Power BI, improving CX strategy alignment.

Insurance Cost Predictor + Customer Segmentation App

- Developed an insurance cost prediction tool using a Random Forest model trained on demographic and health data of over 130K records.
- Integrated KMeans segmentation for profiling customers into high, medium, and low-risk tiers.
- Deployed on Streamlit with histogram visualizations and dynamic risk band feedback for real-time stakeholder use.

CERTIFICATES

- Google Cloud Certified Associate Cloud Engineer
- Python for Data Science – IBM / Cognitive Class (Course ID: PY0101EN)
- Generative AI with Large Language Models - DeepLearning.AI & OpenAI